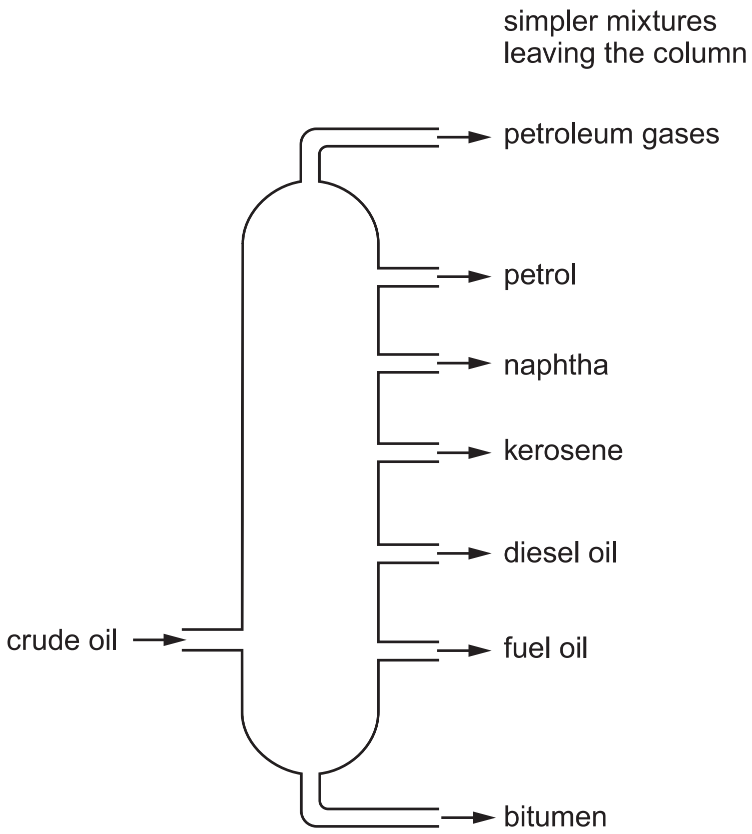
Examiner only

Answer **all** questions.

1. (a) The diagram shows a fractionating column where crude oil is separated into simpler mixtures.



(i) Underline the correct word in the brackets to complete each sentence about the separation process.

I. Before the crude oil mixture enters the column it must be (**boiled / melted / filtered**). [1]

II. Inside the column, the simpler mixtures (**evaporate / condense / solidify**). [1]

III. The simpler mixtures that leave the fractionating column are called (**monomers / polymers / fractions**). [1]

IV. The reason the simpler mixtures are separated from one another is that they have different (**masses / densities / boiling points**). [1]

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(ii) The table shows some information about the simpler mixtures obtained from crude oil.

Simpler mixture	Number of carbon atoms in the compounds	Use
petroleum gases	1–4	cooking
petrol	4–12	cars
naphtha	7–14	making chemicals
kerosene	11–15	aircraft
diesel oil	15–19	lorries
fuel oil	20–30	power stations
bitumen	more than 30	roads

I. Give the name of a mixture **not** used as a fuel. [1]

....

II. Give the names of the **two** mixtures that contain compounds with **13 carbon** atoms.

.... and .... [1]